

The Standard Quantum Limit in Measurement Physics

A Scaffolded Treatise

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Abstract

The standard quantum limit (SQL) denotes several precision benchmarks whose form depends on the measurement model. In continuous position sensing, it arises from optimizing the competition between readout imprecision and quantum backaction for an uncorrelated meter. In phase estimation, it describes the $N^{-1/2}$ uncertainty scaling of independent probes, whereas in phase-preserving amplification it constrains the added noise required to preserve bosonic commutators. This article develops these limits using calibrated measurement records, spectral densities, linear response theory, and quantum estimation theory, then relates their underlying assumptions within a common measurement framework. Particular attention is given to the distinction among readout imprecision, added measurement noise, and total measured sensitivity. This distinction determines how a reported sub-SQL result should be interpreted, since its physical meaning depends on the chosen estimator, reference limit, bandwidth, spectral convention, and noise terms included in the comparison. The final sections show how probe preparation, engineered correlations, and the choice of observable alter the measurement assumptions from which each benchmark follows.

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1 Scope, terminology, and historical development

The following assumes familiarity with expectation values, variances, canonical commutators, and elementary Fourier transforms. Density matrices and Fisher information are introduced in later sections before they enter the analysis.

Precision measurement begins with a physical parameter (θ), such as displacement, force, strain, phase, frequency, or magnetic field, whose influence is encoded in a responsive physical system. Depending on the experiment, that response may appear as the motion of a mirror, a shift in a resonator, a change in an optical or microwave mode, or the collective evolution of an atomic or spin ensemble. A transduction mechanism then converts the physical response into an observable classical record, such as a voltage trace or a sequence of photon counts. Finally, a calibrated estimator interprets that record to produce an estimate ($\hat{\theta}$) together with its associated uncertainty.

The meaning of the term standard quantum limit is determined by the measurement chain, the resources available to it, and the adopted noise conventions. In a conventional continuous position measurement, the SQL arises from the balance between measurement imprecision and quantum backaction. Quantum metrology is the field that studies how quantum states, interactions, and measurement strategies determine the precision with which physical parameters can be estimated. Within this framework, an ensemble of N independent probes gives the statistical SQL, under which the estimation uncertainty decreases as $N^{-1/2}$. For a phase-preserving amplifier, a related quantum limit constrains the added noise required to preserve bosonic commutation relations. These limits may coexist within the same laboratory system, although a meaningful numerical comparison requires consistent definitions of the measured quantity, estimator, bandwidth, resource count, and spectral convention [1–5].

Table 1: Four uses of quantum limit language that require separate assumptions.

Setting	Bounded quantity	Assumptions that produce the standard benchmark	Common expression
Continuous position meter	Added displacement or force spectrum	Linear response, minimum uncertainty meter, and zero useful imprecision and backaction correlation	$\bar{S}_{xx}^{\text{SQL}} = \hbar \chi_m $
Statistical phase estimation	Estimator standard deviation	Independent probes, additive generator, and fixed single probe resource	$\Delta\theta \propto N^{-1/2}$
Phase preserving amplifier	Input referred added noise	Deterministic linear gain for both field quadratures	$n_{\text{add}} \geq 1/2$ at large gain
Correlation aware force sensing	Added force spectrum	Stationary linear sensing with optimized correlations and dissipative response	$\bar{S}_{FF}^{\text{add}} \geq \hbar \text{Im } \chi_m^{-1} $

The early theory of quantum limited measurement asked how a weak force could be inferred from a macroscopic oscillator after the measuring apparatus itself was treated as a quantum system. Braginskii showed that measurement disturbance belongs inside the sensitivity calculation [1]. Work with Vorontsov and Manukin then developed protocols that preserved selected observables across repeated measurements, which led to the theory of quantum nondemolition measurement [6, 7]. This approach became especially important for gravitational wave detection, where repeated interrogation must retain information about an extremely weak mechanical response [2, 8, 9].

Caves carried the same problem into optical interferometry by separating photon counting imprecision from radiation pressure backaction. Squeezed input altered their balance, and his amplifier analysis showed that commutator preservation imposes an added noise cost on phase preserving gain [10–12]. Later linear response and optomechanical treatments expressed these results in terms of detector susceptibility, spectral covariance, and exploitable correlations, which provides the framework used throughout this article [3, 13–17]. Spatial estimation extended the framework to imaging, where the parameter is encoded in the modal structure of the optical field [18].

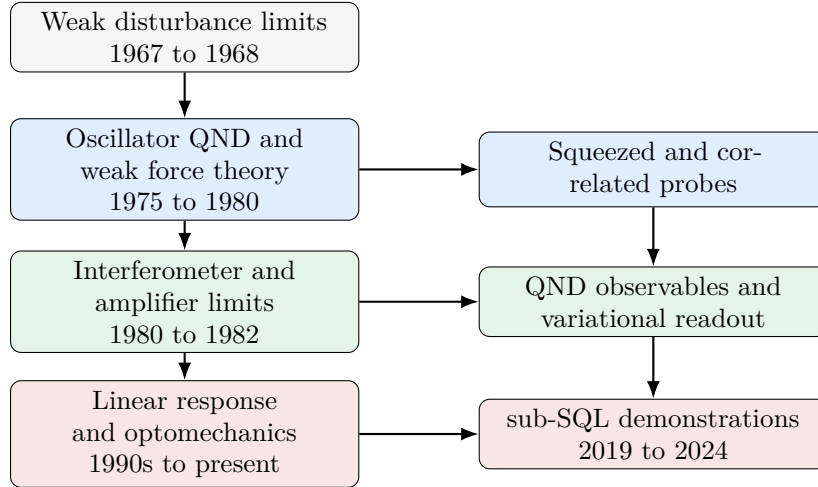


Figure 1: Conceptual lineage of the SQL. The arrows indicate dependence among ideas and carry no claim of exclusive authorship.

Table 2: Core vocabulary used throughout the article.

Term	Meaning
Imprecision	Noise referred to the input that obscures the measured variable in the readout record. Optical phase fluctuations and amplifier output noise are common sources.
Backaction	Fluctuating disturbance exerted by the meter on the measured system. Radiation pressure force and charge force noise are common examples.
Susceptibility	Complex linear response $\chi_m(\Omega)$ that maps applied force to displacement. Its magnitude sets signal gain, and its phase sets the correlation needed for cancellation.
Spectral density	Frequency resolved covariance of stationary fluctuations. Its numerical value depends on sidedness, symmetrization, and Fourier normalization.
Coherent state	Minimum uncertainty optical state whose two quadratures have equal vacuum variance. It supplies the common laser reference.
Squeezed state	State with reduced variance in one field quadrature and increased variance in its conjugate quadrature.
Quantum nondemolition measurement	Repeated measurement of an observable whose future values remain compatible with earlier measurements under the relevant dynamics and coupling.
Backaction evasion	Coupling or readout design that directs disturbance away from the signal observable or cancels its contribution in the estimator.
Quantum Fisher information	Largest local Fisher information available from a parameter dependent quantum state after optimization over quantum measurements.

Practical rule of thumb

Write the measured variable, resource, probe state, readout, bandwidth, and spectral convention beside every SQL formula. An unlabeled SQL curve carries incomplete physical information.

2 Measurement records, response functions, and spectra

The following assumes familiarity with complex Fourier amplitudes and the linear response of a driven harmonic oscillator. The symmetrized spectral density is introduced below, so no previous experience with quantum noise spectra is required.

2.1 From a physical disturbance to an estimator

A linear sensor can be understood as a chain of transduction steps. An external force perturbs a mechanical coordinate, whose response is acquired by an optical or electrical probe. The detector converts the probe response into a classical signal, and a filter processes that signal to estimate the applied force or another parameter of interest. Every stage contributes gain, bandwidth, loss, and noise. Expressing these contributions in terms of a common input variable allows their fluctuations and correlations to be combined within a single covariance model.

Let $x(t)$ denote the displacement of a mechanical mode and $F_{\text{sig}}(t)$ the force to be estimated. The Fourier domain response is

$$x(\Omega) = \chi_m(\Omega) [F_{\text{sig}}(\Omega) + F_{\text{ba}}(\Omega) + F_{\text{th}}(\Omega)], \quad (1)$$

where F_{ba} is meter backaction and F_{th} represents environmental force noise. For an oscillator with effective mass m , resonance Ω_m , and energy damping rate Γ_m ,

$$\chi_m(\Omega) = \frac{1}{m(\Omega_m^2 - \Omega^2 - i\Gamma_m\Omega)}. \quad (2)$$

The free mass approximation follows from $\Omega_m = 0$ and negligible damping over the signal band, which gives $\chi_m = -1/(m\Omega^2)$.

After calibration to displacement, the measured record is

$$y(\Omega) = x(\Omega) + x_{\text{imp}}(\Omega), \quad (3)$$

where x_{imp} is input referred imprecision. Equations (1) and (3) form the minimal model behind the continuous measurement SQL.

2.2 Symmetrized double-sided spectra

For stationary Hermitian fluctuations $\delta\hat{A}$ and $\delta\hat{B}$, this article uses

$$\bar{S}_{AB}(\Omega) = \frac{1}{2} \int_{-\infty}^{\infty} dt e^{i\Omega t} \left\langle \left\{ \delta\hat{A}(t), \delta\hat{B}(0) \right\} \right\rangle. \quad (4)$$

The bar denotes a symmetrized double-sided spectrum per unit angular frequency. For an auto-spectrum, or for an even classical symmetrized spectrum, the corresponding one-sided spectrum defined for $\Omega > 0$ is

$$S_{AB}^{(1)}(\Omega) = 2\bar{S}_{AB}(\Omega). \quad (5)$$

An amplitude spectral density is the square root of a power spectral density. Displacement amplitude spectra carry units of $\text{m}/\sqrt{\text{Hz}}$, force amplitude spectra carry $\text{N}/\sqrt{\text{Hz}}$, and strain amplitude spectra carry $1/\sqrt{\text{Hz}}$ after conversion to spectra per hertz.

Convention checkpoint

Every factor of two in an SQL spectrum depends on sidedness and symmetrization. Equations in this article use $\bar{S}$ for symmetrized double-sided spectra. A one sided result follows from Eq. (5).

At each angular frequency, the meter noise is characterized by the spectral density matrix

$$\bar{\mathbf{S}}_m(\Omega) = \begin{pmatrix} \bar{S}_{xx}^{\text{imp}} & \bar{S}_{xF} \\ \bar{S}_{xF}^* & \bar{S}_{FF}^{\text{ba}} \end{pmatrix}. \quad (6)$$

Referring the measurement record to an equivalent oscillator displacement yields the added displacement noise spectrum

$$\bar{S}_{xx}^{\text{add}}(\Omega) = \bar{S}_{xx}^{\text{imp}} + |\chi_m(\Omega)|^2 \bar{S}_{FF}^{\text{ba}} + 2 \text{Re}[\chi_m^*(\Omega) \bar{S}_{xF}]. \quad (7)$$

After all noise contributions have been expressed in displacement units, the measured spectrum is

$$\bar{S}_{xx}^{\text{meas}} = \bar{S}_{xx}^{\text{intrinsic}} + \bar{S}_{xx}^{\text{add}} + \bar{S}_{xx}^{\text{tech}}, \quad (8)$$

where $\bar{S}_{xx}^{\text{intrinsic}}$ contains the equilibrium mechanical motion, including its thermal and zero-point contributions.

The cross spectrum $\bar{S}_{xF}$ between measurement imprecision and backaction controls the correlation contribution to the added noise. A phase quadrature readout of a coherent probe often provides little exploitable correlation. Variational homodyne detection can instead select a readout quadrature that exploits correlations associated with ponderomotive squeezing, allowing the correlation term in Eq. (7) to become negative over a selected frequency range and thereby reduce the added displacement noise.

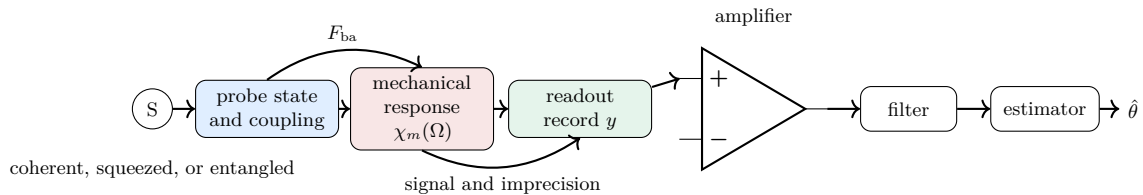


Figure 2: A calibrated quantum measurement chain. SQL performance depends on the full chain and on correlations retained by the estimator.

Practical rule of thumb

Refer every noise source to one variable before addition. Preserve the cross spectral terms until the chosen estimator and readout quadrature have been fixed.

3 Derivation of the continuous position and force SQL

Review Eqs. (2), (6), and (7). The derivation uses a minimum uncertainty linear meter with negligible reverse gain and states every spectral convention explicitly.

3.1 Quantum noise constraint

A stationary linear detector couples the measured coordinate $\hat{x}$ to a detector force operator $\hat{F}$ and records an output operator $\hat{I}$. We adopt the interaction Hamiltonian and linear response relations

$$\hat{H}_{\text{int}} = -\hat{x}\hat{F}, \quad \hat{I}(\Omega) = \hat{I}_0(\Omega) + \chi_{IF}(\Omega)\hat{x}(\Omega), \quad \hat{F}(\Omega) = \hat{F}_0(\Omega) + \chi_{FF}(\Omega)\hat{x}(\Omega). \quad (9)$$

Here $\hat{I}_0$ and $\hat{F}_0$ denote the detector fluctuations evaluated without coupling to the measured system. The susceptibility convention consistent with Eq. (9) is

$$\chi_{AB}(t) = \frac{i}{\hbar} \Theta(t) \langle [\hat{A}_0(t), \hat{B}_0(0)] \rangle. \quad (10)$$

Forward susceptibility χ_{IF} converts the measured coordinate into an output signal. Meanwhile, the reverse susceptibility χ_{FI} describes the response at the detector input port to a perturbation applied at its output, and χ_{FF} describes dynamical backaction, including detector induced damping and frequency shifts.

All spectral densities below are symmetrized, double sided, and evaluated at the same angular frequency Ω . Define the gain asymmetry

$$\tilde{\chi}_{IF}(\Omega) = \chi_{IF}(\Omega) - \chi_{FI}^*(\Omega). \quad (11)$$

The general finite frequency quantum noise constraint is

$$\bar{S}_{II}\bar{S}_{FF} - |\bar{S}_{IF}|^2 \geq \left| \frac{\hbar\tilde{\chi}_{IF}}{2} \right|^2 \left[1 + \Delta\left(\frac{\bar{S}_{IF}}{\hbar\tilde{\chi}_{IF}/2} \right) \right], \quad (12)$$

where

$$\Delta[z] = \frac{|1+z^2| - (1+|z|^2)}{2}, \quad -1 \leq \Delta[z] \leq 0. \quad (13)$$

Here, the function $\Delta[z]$ retains phase information carried by a complex output-backaction cross spectrum. It vanishes whenever z is real and therefore vanishes in the uncorrelated case $\bar{S}_{IF} = 0$ [3].

A detector output that produces a classical continuous record must also be simultaneously measurable at distinct times,

$$[\hat{I}(t), \hat{I}(t')] = 0. \quad (14)$$

Within the stationary linear detector model, this condition gives $\chi_{II} = \chi_{FI} = 0$. Retaining the force self susceptibility χ_{FF} then yields the tighter finite frequency relation

$$\bar{S}_{II}\bar{S}_{FF} - |\bar{S}_{IF}|^2 \geq \frac{\hbar^2}{4} |\chi_{IF}|^2 + \hbar \left| \text{Im} \left[\bar{S}_{IF}^* \chi_{IF} - \chi_{FF} \bar{S}_{II} \right] \right|. \quad (15)$$

Input referral divides the output fluctuation by the complex forward gain,

$$\hat{x}_{\text{imp}}(\Omega) = \frac{\hat{I}_0(\Omega)}{\chi_{IF}(\Omega)}, \quad \bar{S}_{xx}^{\text{imp}} = \frac{\bar{S}_{II}}{|\chi_{IF}|^2}, \quad \bar{S}_{xF} = \frac{\bar{S}_{IF}}{\chi_{IF}}, \quad \bar{S}_{FF}^{\text{ba}} = \bar{S}_{FF}. \quad (16)$$

Dividing Eq. (15) by $|\chi_{IF}|^2$ gives

$$\bar{S}_{xx}^{\text{imp}} \bar{S}_{FF}^{\text{ba}} - |\bar{S}_{xF}|^2 \geq \frac{\hbar^2}{4} + \hbar \left| \text{Im} \left[\bar{S}_{xF}^* - \chi_{FF} \bar{S}_{xx}^{\text{imp}} \right] \right|. \quad (17)$$

The second term on the right is nonnegative, so every detector satisfying these continuous output conditions obeys the necessary determinant bound

$$\bar{S}_{xx}^{\text{imp}} \bar{S}_{FF}^{\text{ba}} - |\bar{S}_{xF}|^2 \geq \frac{\hbar^2}{4}. \quad (18)$$

Equality for an ideal detector also requires the phase condition

$$\text{Im} \bar{S}_{xF} = -\text{Im} \chi_{FF} \bar{S}_{xx}^{\text{imp}}. \quad (19)$$

Consequently, Eq. (18) is a necessary uncertainty constraint, whereas Eqs. (17) and (19) retain the additional information required to establish physical realizability at finite frequency [19, 20].

The SQL derivation in the following subsection uses an uncorrelated meter, $\bar{S}_{xF} = 0$, and neglects detector induced damping, $\text{Im} \chi_{FF} \simeq 0$. A minimum uncertainty detector can then saturate Eq. (18), giving

$$\bar{S}_{xx}^{\text{imp}} \bar{S}_{FF}^{\text{ba}} = \frac{\hbar^2}{4}. \quad (20)$$

This relation supplies the imprecision–backaction tradeoff used to derive the conventional uncorrelated continuous measurement SQL.

3.2 Optimization for an uncorrelated meter

Consider one analysis frequency Ω and set $\bar{S}_{xF} = 0$. Let the positive dimensionless parameter λ represent measurement strength, and write

$$\bar{S}_{xx}^{\text{imp}}(\Omega) = \frac{A(\Omega)}{\lambda}, \quad \bar{S}_{FF}^{\text{ba}}(\Omega) = B(\Omega)\lambda, \quad (21)$$

where $A(\Omega) > 0$ and $B(\Omega) > 0$ are independent of λ . Their frequency arguments will be suppressed during the pointwise optimization. Increasing λ extracts more information from the measured coordinate, which reduces imprecision as λ^{-1} , and increases the fluctuating backaction force in proportion to λ .

With zero imprecision and backaction correlation, Eq. (7) reduces to

$$\bar{S}_{xx}^{\text{add}}(\Omega, \lambda) = \frac{A}{\lambda} + |\chi_m(\Omega)|^2 B \lambda. \quad (22)$$

The two terms are respectively the input referred imprecision and the displacement produced by backaction. Differentiation gives

$$\frac{\partial \bar{S}_{xx}^{\text{add}}}{\partial \lambda} = -\frac{A}{\lambda^2} + |\chi_m|^2 B, \quad \frac{\partial^2 \bar{S}_{xx}^{\text{add}}}{\partial \lambda^2} = \frac{2A}{\lambda^3} > 0. \quad (23)$$

The stationary point is therefore the unique minimum,

$$\lambda_{\text{opt}}(\Omega) = \sqrt{\frac{A(\Omega)}{|\chi_m(\Omega)|^2 B(\Omega)}}. \quad (24)$$

Substitution into the two noise terms shows that they contribute equally,

$$\frac{A}{\lambda_{\text{opt}}} = |\chi_m|^2 B \lambda_{\text{opt}} = |\chi_m| \sqrt{AB}. \quad (25)$$

Their sum is

$$\bar{S}_{xx,\text{min}}^{\text{add}}(\Omega) = 2|\chi_m(\Omega)|\sqrt{A(\Omega)B(\Omega)}. \quad (26)$$

For the minimum uncertainty uncorrelated meter introduced in the preceding subsection,

$$A(\Omega)B(\Omega) = \bar{S}_{xx}^{\text{imp}}(\Omega)\bar{S}_{FF}^{\text{ba}}(\Omega) = \frac{\hbar^2}{4}. \quad (27)$$

Equation (26) then gives the conventional uncorrelated continuous position SQL,

$$\boxed{\bar{S}_{xx}^{\text{SQL}}(\Omega) = \hbar|\chi_m(\Omega)|}. \quad (28)$$

The equivalent force referred added spectrum is defined by

$$\bar{S}_{FF}^{\text{add}}(\Omega) = \frac{\bar{S}_{xx}^{\text{add}}(\Omega)}{|\chi_m(\Omega)|^2}. \quad (29)$$

Consequently,

$$\boxed{\bar{S}_{FF}^{\text{SQL}}(\Omega) = \frac{\hbar}{|\chi_m(\Omega)|}}. \quad (30)$$

The normalized strength

$$q = \frac{\lambda}{\lambda_{\text{opt}}} \quad (31)$$

places the three noise curves in a universal form,

$$\frac{\bar{S}_{xx}^{\text{imp}}}{\bar{S}_{xx}^{\text{SQL}}} = \frac{1}{2q}, \quad \frac{|\chi_m|^2 \bar{S}_{FF}^{\text{ba}}}{\bar{S}_{xx}^{\text{SQL}}} = \frac{q}{2}, \quad \frac{\bar{S}_{xx}^{\text{add}}}{\bar{S}_{xx}^{\text{SQL}}} = \frac{1}{2} \left(q + \frac{1}{q} \right) \geq 1. \quad (32)$$

Equality occurs at $q = 1$, where imprecision and backaction each contribute one half of the SQL. Correlated meters and measurements of different observables can have optima that do not satisfy this equal contribution condition.

3.3 Free mass and resonant oscillator

For a free mass, the susceptibility is

$$\chi_m(\Omega) = -\frac{1}{m\Omega^2}, \quad |\chi_m(\Omega)| = \frac{1}{m\Omega^2}. \quad (33)$$

The displacement and force referred SQL spectra are therefore

$$\bar{S}_{xx}^{\text{SQL}}(\Omega) = \frac{\hbar}{m\Omega^2}, \quad \bar{S}_{FF}^{\text{SQL}}(\Omega) = \hbar m\Omega^2. \quad (34)$$

For the viscously damped oscillator defined in Eq. (??), evaluation at $\Omega = \Omega_m$ gives

$$\chi_m(\Omega_m) = \frac{i}{m\Gamma_m\Omega_m}, \quad |\chi_m(\Omega_m)| = \frac{1}{m\Gamma_m\Omega_m}. \quad (35)$$

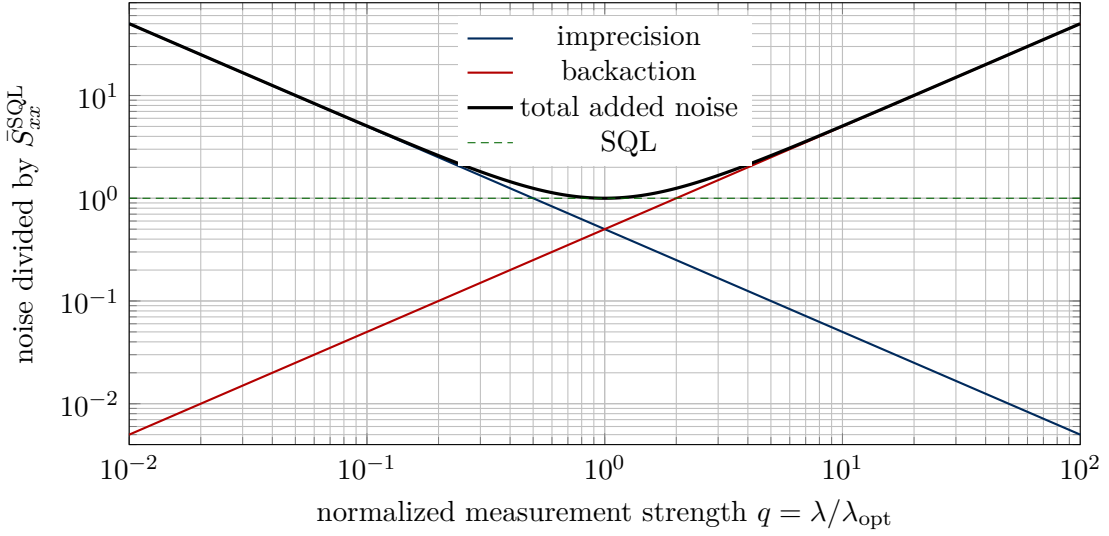


Figure 3: Pointwise imprecision and backaction optimization for an uncorrelated minimum uncertainty linear meter. Every curve is divided by $\bar{S}_{xx}^{\text{SQL}}$, so the total reaches unity at $\lambda = \lambda_{\text{opt}}$.

The corresponding SQL spectra are

$$\bar{S}_{xx}^{\text{SQL}}(\Omega_m) = \frac{\hbar}{m\Gamma_m\Omega_m}, \quad \bar{S}_{FF}^{\text{SQL}}(\Omega_m) = \hbar m\Gamma_m\Omega_m. \quad (36)$$

These expressions use symmetrized double sided spectra per unit angular frequency. For $\Omega > 0$, the associated one sided power spectra are twice the displayed values, and their square root amplitude spectra acquire a factor $\sqrt{2}$.

The symmetrized equilibrium bath force spectrum, including its zero point contribution, has the one sided form

$$S_{FF}^{\text{th},(1)}(\Omega) = 2m\Gamma_m\hbar\Omega \coth\left(\frac{\hbar\Omega}{2k_BT}\right), \quad \Omega > 0. \quad (37)$$

In the classical regime $\hbar\Omega \ll k_BT$,

$$\coth\left(\frac{\hbar\Omega}{2k_BT}\right) \simeq \frac{2k_BT}{\hbar\Omega}, \quad (38)$$

which gives

$$S_{FF}^{\text{th},(1)} \simeq 4m\Gamma_mk_BT. \quad (39)$$

Quantum measurement noise becomes experimentally resolvable as thermal, laser, electronic, and calibration noise are reduced to the scale of the relevant SQL, with every contribution evaluated over the same bandwidth and using the same spectral convention. [3].

3.4 Frequency dependent optimum

The preceding minimization is pointwise in frequency. A single fixed probe strength generally reaches its optimum across a limited interval because the mechanical susceptibility varies with Ω . Introduce

$$r = \frac{\Omega}{\Omega_m}, \quad \gamma = \frac{\Gamma_m}{\Omega_m}, \quad \chi_0 = \frac{1}{m\Omega_m^2}. \quad (40)$$

The oscillator susceptibility can then be written

$$\chi_m(\Omega) = \chi_0 \tilde{\chi}_m(r), \quad \tilde{\chi}_m(r) = \frac{1}{1 - r^2 - i\gamma r}, \quad (41)$$

with magnitude

$$|\tilde{\chi}_m(r)| = \frac{1}{\sqrt{(1 - r^2)^2 + (\gamma r)^2}}. \quad (42)$$

For the analytic heat map, take A and B to be frequency independent across the displayed interval. Define a fixed reference strength and its dimensionless ratio by

$$\lambda_0 = \sqrt{\frac{A}{B\chi_0^2}}, \quad p = \frac{\lambda}{\lambda_0}. \quad (43)$$

Dividing Eq. (22) by $\chi_0\sqrt{AB}$ gives the dimensionless added noise

$$N(r, p) \equiv \frac{\bar{S}_{xx}^{\text{add}}(r, p)}{\chi_0\sqrt{AB}} = p^{-1} + p|\tilde{\chi}_m(r)|^2. \quad (44)$$

For a minimum uncertainty meter with $\sqrt{AB} = \hbar/2$, the SQL in these units is

$$N_{\text{SQL}}(r) = 2|\tilde{\chi}_m(r)|. \quad (45)$$

Setting

$$u = p|\tilde{\chi}_m(r)| \quad (46)$$

gives

$$\frac{\bar{S}_{xx}^{\text{add}}}{\bar{S}_{xx}^{\text{SQL}}} = \frac{N}{N_{\text{SQL}}} = \frac{1}{2} (u + u^{-1}) \geq 1. \quad (47)$$

Equality occurs at the frequency dependent optimum

$$p_{\text{opt}}(r) = \frac{1}{|\tilde{\chi}_m(r)|}. \quad (48)$$

A fixed value of p therefore intersects the optimum valley over a limited frequency range.

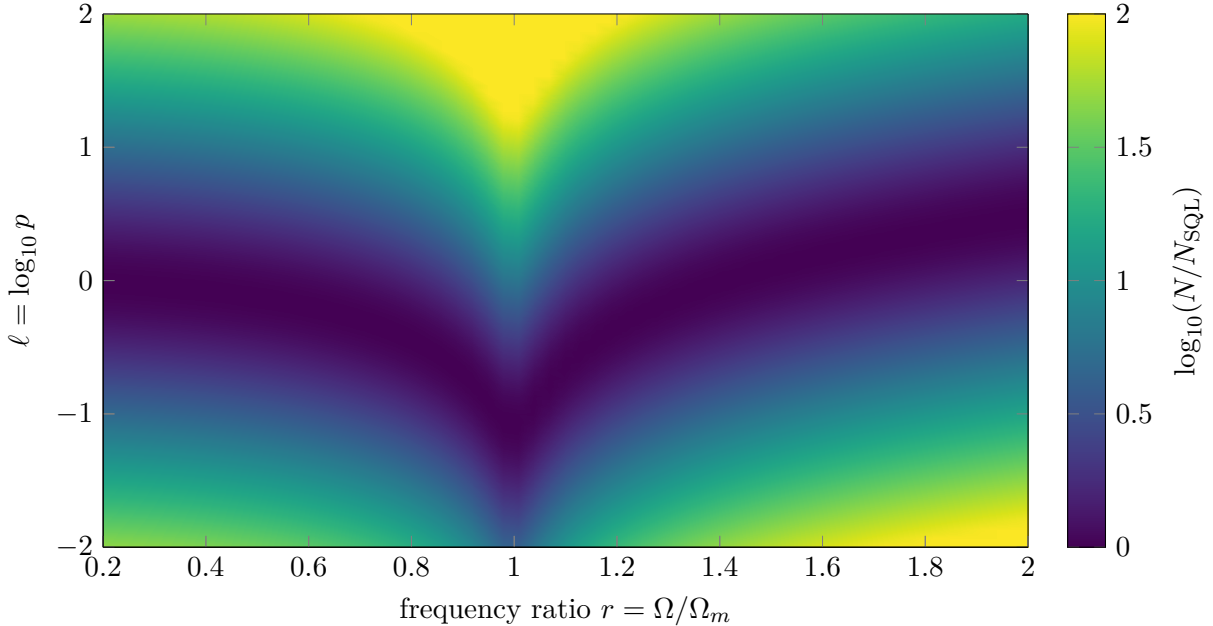


Figure 4: Added displacement noise relative to the local uncorrelated SQL for damping ratio $\gamma = 0.08$. The minimum follows $p_{\text{opt}} = 1/|\tilde{\chi}_m|$. The plotted quantity is evaluated analytically from Eqs. (42) and (47).

3.5 SQL and the correlation-aware lower bound

A complex imprecision and backaction cross spectrum can reduce the added noise in Eq. (7), since

$$2 \operatorname{Re} [\chi_m^*(\Omega) \bar{S}_{xF}(\Omega)] \quad (49)$$

can be negative. For a conventional stationary linear position meter with the bilinear coupling $\hat{H}_{\text{int}} = -\hat{x}\hat{F}$, optimization over physically admissible correlations gives the correlation-aware displacement bound

$$\bar{S}_{xx}^{\text{add}}(\Omega) \geq \hbar |\operatorname{Im}[\chi_m(\Omega)]|. \quad (50)$$

Force referral using Eq. (29) gives

$$\bar{S}_{FF}^{\text{add}}(\Omega) \geq \hbar |\operatorname{Im}[\chi_m^{-1}(\Omega)]|, \quad (51)$$

since

$$\frac{|\operatorname{Im} \chi_m|}{|\chi_m|^2} = |\operatorname{Im}[\chi_m^{-1}]|. \quad (52)$$

Equations (50) and (51) are therefore the displacement and force referred forms of the same bound.

For the viscously damped oscillator,

$$\chi_m^{-1}(\Omega) = m (\Omega_m^2 - \Omega^2 - i\Gamma_m \Omega), \quad (53)$$

so

$$|\operatorname{Im}[\chi_m^{-1}(\Omega)]| = m\Gamma_m |\Omega|. \quad (54)$$

The force referred bound becomes the dissipative quantum limit,

$$\boxed{\bar{S}_{FF}^{\text{DQL}}(\Omega) = \hbar m \Gamma_m |\Omega|}. \quad (55)$$

The DQL constrains the optimized meter-added force noise. Equation (37) describes the distinct equilibrium force noise associated with the probe bath, although the two expressions have the same zero temperature spectral scale.

For the oscillator model used here, $\chi_m(\Omega_m)$ is purely imaginary. The uncorrelated SQL and the dissipative bound therefore coincide exactly at resonance. Their ratio away from resonance is

$$\frac{\bar{S}_{xx}^{\text{DQL}}}{\bar{S}_{xx}^{\text{SQL}}} = \frac{|\operatorname{Im} \chi_m|}{|\chi_m|} = \frac{\gamma|r|}{\sqrt{(1-r^2)^2 + (\gamma r)^2}} \leq 1. \quad (56)$$

The separation between the two curves quantifies the pointwise improvement available from optimized correlations within this stationary measurement model.

For an ideal lossless free mass, $\operatorname{Im} \chi_m^{-1} = 0$, so the stationary dissipative bound supplies no finite sensitivity scale. Finite observation duration, available probe energy, optical loss, causal filtering, and the chosen signal waveform then supply additional constraints. More general detector and probe interactions can also produce generalized bounds outside the scope of Eqs. (50) and (51) [21–23].

Practical rule of thumb

Balance imprecision and backaction to locate the uncorrelated SQL. Preserve the complex cross spectrum when correlations enter the estimator, and use the dissipative part of the probe response to evaluate the conventional stationary correlation-aware bound.

4 Optical interferometers and cavity optomechanics

This section assumes familiarity with optical phase, coherent state photon statistics, and the susceptibility of a harmonic oscillator. The spectral relations needed for the discussion were developed in Section 2.

4.1 Coherent light in a Michelson position meter

A differential arm displacement x changes the round trip phase by

$$\phi = 2k_0 x = \frac{4\pi x}{\lambda_0}. \quad (57)$$

For N detected photons in a coherent state, phase imprecision scales as

$$\Delta\phi_{\text{imp}} \sim N^{-1/2}, \quad \Delta x_{\text{imp}} \sim \frac{\lambda_0}{4\pi\sqrt{N}}. \quad (58)$$

With optical power P , integration time τ , and carrier frequency ω_0 , the detected photon number scales as $N \sim P\tau/(\hbar\omega_0)$. The displacement imprecision spectrum therefore scales as P^{-1} .

Coherent state amplitude quadrature fluctuations produce photon flux fluctuations around a bright carrier. Their fluctuating momentum transfer produces a radiation pressure force spectrum proportional to P . The phase quadrature produces readout imprecision. The equivalent displacement spectrum from force noise is multiplied by $|\chi_m|^2$, so a compact model is

$$\bar{S}_{xx}^{\text{add}}(\Omega; P) = \frac{\alpha(\Omega)}{P} + \beta(\Omega)P|\chi_m(\Omega)|^2. \quad (59)$$

The optical commutator fixes $\alpha\beta$ for an ideal coherent probe. Minimization over P reproduces Eq. (28). Strain calibration in a simple differential arm model uses $x = hL$, which gives $S_{hh} = S_{xx}/L^2$ after the effective differential mechanical mode has been defined.

The full interferometer response is shaped by optical storage in the arm and signal recycling cavities, together with the selected homodyne quadrature and the losses accumulated before detection. These elements rotate and attenuate the output noise ellipse in a frequency dependent manner. A filter cavity applies a corresponding rotation to the injected squeezed state, aligning its reduced noise quadrature with phase readout at high frequency and with radiation pressure suppression at low frequency [24–26].

4.2 Cavity optomechanical implementation

For a single optical and mechanical mode, the dispersive Hamiltonian is

$$\hat{H} = \hbar\omega_c \hat{a}^\dagger \hat{a} + \hbar\Omega_m \hat{b}^\dagger \hat{b} - \hbar g_0 \hat{a}^\dagger \hat{a} (\hat{b} + \hat{b}^\dagger). \quad (60)$$

The single photon coupling g_0 gives the cavity frequency shift produced by zero point motion. Driving the cavity to mean photon number n_c produces the linearized coupling $G = g_0\sqrt{n_c}$. For a resonant drive in the unresolved sideband measurement limit, a representative measurement rate is

$$\Gamma_{\text{meas}} \simeq \frac{4\eta G^2}{\kappa}, \quad (61)$$

where η is total detection efficiency and κ is the cavity energy decay rate. Larger n_c increases information rate and radiation pressure shot noise together. The cooperativity

$$C = \frac{4G^2}{\kappa\Gamma_m} \quad (62)$$

compares optomechanical measurement dynamics with mechanical damping. Quantum cooperativity also accounts for thermal occupation and indicates whether quantum backaction can compete with thermalization [16, 27].

The oscillator zero point displacement is

$$x_{\text{zpf}} = \sqrt{\frac{\hbar}{2m\Omega_m}}. \quad (63)$$

For $k_B T \gg \hbar\Omega_m$, the thermal occupation is $n_{\text{th}} \simeq k_B T / (\hbar\Omega_m)$. The product $\Gamma_m n_{\text{th}}$ estimates a thermal heating rate. At fixed quality factor $Q = \Omega_m / \Gamma_m$,

$$\Gamma_m n_{\text{th}} \simeq \frac{k_B T}{\hbar Q}. \quad (64)$$

Frequency alone therefore gives no reduction of this rate at fixed Q . Material loss, temperature, surface contamination, optical absorption, and mode shape determine the realized thermal environment.

Experiments have observed radiation pressure backaction, measured forces close to the SQL, and used optomechanical correlations to obtain total off resonant force and displacement sensitivity below the uncorrelated SQL [21, 28, 29]. The last statement is stronger than a report of imprecision below an SQL scale. It includes the calibrated intrinsic mechanical motion, measurement imprecision, backaction, and their correlations in the measured band.

Practical rule of thumb

Increasing optical power improves phase readout until radiation pressure, absorption, heating, or dynamical backaction erases the gain. Track detected efficiency, intracavity power, and mechanical thermalization as separate control variables.

5 From continuous records to Fisher information

Density matrices ρ are used as positive, unit trace operators that describe pure and mixed quantum states. The discussion also uses conditional probabilities, expectation values, and variances. The score function, Fisher information, positive operator valued measurements, and matched filtering are defined before they enter the analysis.

5.1 Classical estimators and local information

Suppose that a measurement produces a discrete outcome k with conditional probability $p(k|\theta)$, where θ is the parameter to be estimated. The score associated with one outcome is

$$u_\theta(k) = \frac{\partial}{\partial \theta} \ln p(k|\theta). \quad (65)$$

The score measures how strongly the logarithm of the likelihood changes under a small change in θ . Under the usual regularity conditions,

$$\sum_k p(k|\theta) = 1 \implies \sum_k p(k|\theta) u_\theta(k) = \sum_k \partial_\theta p(k|\theta) = 0. \quad (66)$$

Its mean therefore vanishes. The classical Fisher information is its variance,

$$\begin{aligned} F_C(\theta) &= \sum_k p(k|\theta) \left[\frac{\partial}{\partial \theta} \ln p(k|\theta) \right]^2 \\ &= \sum_{k: p(k|\theta) > 0} \frac{[\partial_\theta p(k|\theta)]^2}{p(k|\theta)}. \end{aligned} \quad (67)$$

For a continuous outcome, the corresponding sum is replaced by an integral.

Consider an estimator $\hat{\theta}$ formed from ν statistically independent measurement outcomes. Local unbiasedness at an operating point θ_0 means

$$\langle \hat{\theta} \rangle_{\theta_0} = \theta_0, \quad \left. \frac{\partial}{\partial \theta} \langle \hat{\theta} \rangle_\theta \right|_{\theta=\theta_0} = 1. \quad (68)$$

The total score is the sum of the scores from the ν outcomes. Independence makes its variance equal to $\nu F_C(\theta_0)$. Differentiating the estimator mean and using Eq. (66) gives

$$1 = \left\langle \left(\hat{\theta} - \theta_0 \right) u_{\theta_0}^{(\nu)} \right\rangle, \quad (69)$$

where $u_{\theta_0}^{(\nu)}$ is the total score. The Cauchy–Schwarz inequality then gives

$$1 \leq \text{Var}_{\theta_0}(\hat{\theta}) \text{Var}_{\theta_0}(u_{\theta_0}^{(\nu)}) = \nu F_C(\theta_0) \text{Var}_{\theta_0}(\hat{\theta}). \quad (70)$$

The resulting Cramér–Rao inequality is

$$\text{Var}_{\theta_0}(\hat{\theta}) \geq \frac{1}{\nu F_C(\theta_0)}. \quad (71)$$

This is a local statement about precision near θ_0 . Finite samples can produce estimator bias, nuisance parameters can couple the desired parameter to additional unknown quantities, and periodic parameters can produce equivalent likelihood maxima. Such cases require finite sample, multiparameter, constrained, or Bayesian analyses matched to the inference problem.

A quantum measurement is described by a positive operator valued measure $\{E_k\}$. For a parameter dependent quantum state ρ_θ , the Born rule gives

$$p(k|\theta) = \text{Tr}(\rho_\theta E_k), \quad E_k \succeq 0, \quad \sum_k E_k = \mathbb{I}. \quad (72)$$

Each POVM produces a classical probability distribution and therefore a classical Fisher information $F_C(\theta; \{E_k\})$. The quantum Fisher information is the largest classical Fisher information accessible from the state at the chosen operating point,

$$F_Q(\theta) = \max_{\{E_k\}} F_C(\theta; \{E_k\}). \quad (73)$$

The symmetric logarithmic derivative L_θ is the Hermitian operator defined on the support of ρ_θ by

$$\partial_\theta \rho_\theta = \frac{1}{2} (L_\theta \rho_\theta + \rho_\theta L_\theta). \quad (74)$$

It gives

$$F_Q(\theta) = \text{Tr}(\rho_\theta L_\theta^2) = \text{Tr}[(\partial_\theta \rho_\theta) L_\theta], \quad F_C(\theta; \{E_k\}) \leq F_Q(\theta). \quad (75)$$

For one parameter, a measurement in an eigenbasis of L_{θ_0} reaches the maximum locally under the usual regularity conditions. That measurement can depend on the operating point, so adaptive calibration can be required when θ_0 is initially unknown.

Combining Eq. (71) with $F_C \leq F_Q$ gives the quantum Cramér–Rao bound

$$\text{Var}_{\theta_0}(\hat{\theta}) \geq \frac{1}{\nu F_Q(\theta_0)}. \quad (76)$$

This bound applies to locally unbiased estimation of one parameter. Multiparameter problems introduce a Fisher information matrix, and incompatible optimal measurements can prevent all matrix bounds from being saturated simultaneously [4, 5, 30].

Consider a unitary parameter encoding

$$\rho_\theta = e^{-i\theta \hat{G}} \rho e^{i\theta \hat{G}}, \quad \rho = \sum_j p_j |j\rangle\langle j|. \quad (77)$$

Here θ is a dimensionless phase. For a dimensional physical parameter, the interaction time and coupling coefficient can be incorporated into $\hat{G}$. Differentiation gives

$$\partial_\theta \rho_\theta = -i[\hat{G}, \rho_\theta]. \quad (78)$$

Evaluation in the eigenbasis of ρ gives

$$F_Q[\rho, \hat{G}] = 2 \sum_{j,k: p_j + p_k > 0} \frac{(p_j - p_k)^2}{p_j + p_k} |\langle j | \hat{G} | k \rangle|^2. \quad (79)$$

For a pure state $|\psi_\theta\rangle = e^{-i\theta \hat{G}} |\psi\rangle$,

$$|\partial_\theta \psi_\theta\rangle = -i\hat{G} |\psi_\theta\rangle. \quad (80)$$

The pure state QFI is therefore

$$\begin{aligned} F_Q &= 4 \left[\langle \partial_\theta \psi_\theta | \partial_\theta \psi_\theta \rangle - |\langle \psi_\theta | \partial_\theta \psi_\theta \rangle|^2 \right] \\ &= 4 \left(\langle \hat{G}^2 \rangle_\psi - \langle \hat{G} \rangle_\psi^2 \right) = 4(\Delta \hat{G})_\psi^2. \end{aligned} \quad (81)$$

The generator variance measures the rate at which neighboring pure states separate under the parameter encoding. State preparation improves sensitivity when it increases this separation in a way that survives the available measurement and subsequent loss.

5.2 Independent probes and the statistical SQL

Let a dimensionless phase be encoded locally into N probes according to

$$\hat{U}_\theta = e^{-i\theta \hat{G}}, \quad \hat{G} = \sum_{j=1}^N \hat{g}_j. \quad (82)$$

For a pure product state

$$|\Psi\rangle = \bigotimes_{j=1}^N |\psi_j\rangle, \quad (83)$$

expectation values factorize between different probes. The covariance of $\hat{g}_j$ and $\hat{g}_k$ therefore vanishes for $j \neq k$, which gives

$$(\Delta \hat{G})_{\Psi}^2 = \sum_{j=1}^N (\Delta \hat{g}_j)_{\psi_j}^2, \quad (84)$$

$$F_Q[|\Psi\rangle, \hat{G}] = 4 \sum_{j=1}^N (\Delta \hat{g}_j)_{\psi_j}^2. \quad (85)$$

Define the spectral range of the j th local generator by

$$R_j = g_{j,\max} - g_{j,\min}. \quad (86)$$

The variance of any state supported between these two extremal eigenvalues obeys

$$(\Delta \hat{g}_j)^2 \leq \frac{R_j^2}{4}. \quad (87)$$

The maximum is reached by an equal coherent superposition of eigenstates at the two endpoints. Equations (85) and (87) give

$$F_Q[|\Psi\rangle, \hat{G}] \leq \sum_{j=1}^N R_j^2. \quad (88)$$

Quantum Fisher information is convex in the state. A separable mixed state, which is a probabilistic mixture of product states, therefore obeys the same bound,

$$F_Q[\rho_{\text{sep}}, \hat{G}] \leq \sum_{j=1}^N R_j^2. \quad (89)$$

For identical probes with the same local variance, Eq. (84) gives

$$(\Delta \hat{G})^2 = N(\Delta \hat{g})^2, \quad F_Q \propto N, \quad \Delta\theta \equiv \sqrt{\text{Var}(\hat{\theta})} \propto N^{-1/2}. \quad (90)$$

This square root scaling is the statistical SQL associated with independent or separable probes under a fixed local generator and fixed resource count.

For qubits with

$$\hat{g}_j = \frac{\sigma_z^{(j)}}{2}, \quad (91)$$

the eigenvalues are $\pm 1/2$, so $R_j = 1$. Equation (89) becomes

$$F_Q[\rho_{\text{sep}}, \hat{G}] \leq N. \quad (92)$$

The quantum Cramér–Rao bound then gives

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\nu N}, \quad \Delta\theta \geq \frac{1}{\sqrt{\nu N}}. \quad (93)$$

This is a precise restricted state benchmark and supplies the mathematical meaning of the statistical SQL [31, 32].

For the same generator, the ideal GHZ state is

$$|\text{GHZ}_N\rangle = \frac{|0\rangle^{\otimes N} + |1\rangle^{\otimes N}}{\sqrt{2}}. \quad (94)$$

Its two components are eigenstates of $\hat{G}$ with eigenvalues $+N/2$ and $-N/2$. Consequently,

$$\langle \hat{G} \rangle_{\text{GHZ}} = 0, \quad (\Delta \hat{G})_{\text{GHZ}}^2 = \frac{N^2}{4}. \quad (95)$$

Equation (81) then gives

$$F_Q = N^2, \quad \text{Var}(\hat{\theta}) \geq \frac{1}{\nu N^2}, \quad \Delta\theta \geq \frac{1}{N\sqrt{\nu}}. \quad (96)$$

This Heisenberg scaling assumes coherent local encoding and a resource definition that counts each probe interaction. Sequential coherent reuse can produce the same formal scaling when every passage through the encoding interaction is included in the resource count. Under common independent loss and Markovian dephasing models, the optimal QFI grows asymptotically in proportion to N , so the uncertainty returns to $N^{-1/2}$ scaling and the remaining quantum enhancement appears as a constant prefactor [33–35].

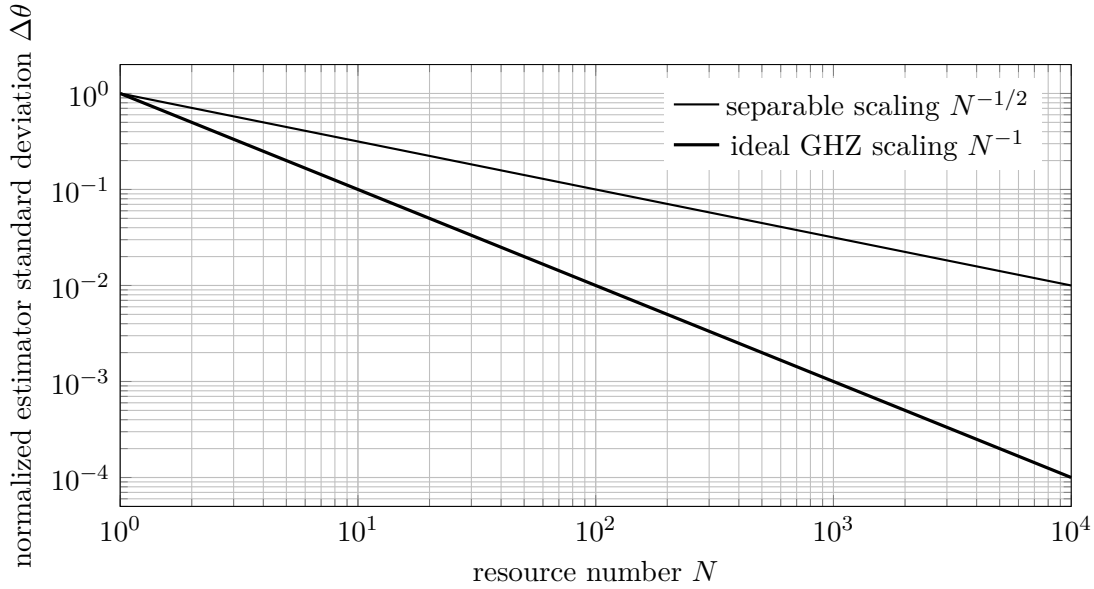


Figure 5: Ideal estimator standard deviation under separable and GHZ phase encoding. Loss, decoherence, interrogation time, and state preparation overhead determine the experimentally realized advantage.

5.3 The bridge from spectra to estimators

The continuous measurement analysis describes a noisy record as a function of time or Fourier frequency. Fisher information describes how strongly the probability law of that complete record changes with the parameter. A stationary Gaussian model makes the connection explicit.

Let a real recorded signal have the form

$$y(\Omega) = \theta s(\Omega) + n(\Omega), \quad (97)$$

where θ is an unknown real amplitude, $s(\Omega)$ is a known signal template that includes the observation window and detector response, and $n(\Omega)$ is zero mean stationary Gaussian noise. Its double sided spectrum is defined by

$$\langle n(\Omega)n^*(\Omega') \rangle = 2\pi\delta(\Omega - \Omega')\bar{S}_{nn}(\Omega). \quad (98)$$

Because $y(t)$ and $n(t)$ are classical records, $\bar{S}_{nn}$ is an ordinary classical covariance spectrum. The bar retains the double sided convention used in the preceding sections.

Define the noise weighted inner product

$$(a|b)_n = \int_{-\infty}^{\infty} \frac{d\Omega}{2\pi} \frac{a^*(\Omega)b(\Omega)}{\bar{S}_{nn}(\Omega)}. \quad (99)$$

For real time domain signals, conjugate symmetry makes the full double sided integral real. If the noise spectrum is locally independent of θ , the Gaussian log likelihood is

$$\ln p[y|\theta] = \text{const.} - \frac{1}{2}(y - \theta s | y - \theta s)_n. \quad (100)$$

Differentiation gives the score of the complete record,

$$\frac{\partial}{\partial \theta} \ln p[y|\theta] = (s | y - \theta s)_n = (s|n)_n. \quad (101)$$

Using Eq. (98), the Fisher information carried by one record is

$$\begin{aligned} F_C^{\text{rec}} &= \left\langle \left[\frac{\partial}{\partial \theta} \ln p[y|\theta] \right]^2 \right\rangle \\ &= (s|s)_n = \int_{-\infty}^{\infty} \frac{d\Omega}{2\pi} \frac{|s(\Omega)|^2}{\bar{S}_{nn}(\Omega)}. \end{aligned} \quad (102)$$

Maximizing Eq. (100), or equivalently minimizing its quadratic exponent, gives

$$0 = \frac{\partial}{\partial \theta} (y - \theta s | y - \theta s)_n = -2(s|y)_n + 2\theta(s|s)_n. \quad (103)$$

The resulting matched filter estimator is

$$\hat{\theta} = \frac{(s|y)_n}{(s|s)_n} = \frac{\int_{-\infty}^{\infty} \frac{d\Omega}{2\pi} \frac{s^*(\Omega)y(\Omega)}{\bar{S}_{nn}(\Omega)}}{\int_{-\infty}^{\infty} \frac{d\Omega}{2\pi} \frac{|s(\Omega)|^2}{\bar{S}_{nn}(\Omega)}}. \quad (104)$$

Substituting $y = \theta s + n$ gives

$$\hat{\theta} - \theta = \frac{(s|n)_n}{(s|s)_n}. \quad (105)$$

Equation (98) then gives

$$\text{Var}(\hat{\theta}) = \frac{1}{(s|s)_n} = \left[\int_{-\infty}^{\infty} \frac{d\Omega}{2\pi} \frac{|s(\Omega)|^2}{\bar{S}_{nn}(\Omega)} \right]^{-1} = \frac{1}{F_C^{\text{rec}}}. \quad (106)$$

The matched filter therefore saturates the classical Cramér–Rao bound for this linear Gaussian amplitude model. For ν independent records, the total Fisher information is νF_C^{rec} .

If the applied force has a known waveform

$$F_{\text{sig}}(\Omega) = \theta F_0(\Omega), \quad (107)$$

then a displacement calibrated record has the template

$$s(\Omega) = \chi_m(\Omega) F_0(\Omega). \quad (108)$$

A strain measurement with $h(\Omega) = \theta h_0(\Omega)$ uses

$$s(\Omega) = R_h(\Omega) h_0(\Omega), \quad (109)$$

where R_h is the calibrated interferometer response.

A measurement control parameter such as probe strength λ can alter the imprecision, backaction, and cross spectrum. Dynamical backaction can also alter the mechanical response and therefore the signal template. The Fisher information to optimize is consequently

$$F_C^{\text{rec}}(\lambda) = \int_{-\infty}^{\infty} \frac{d\Omega}{2\pi} \frac{|s(\Omega; \lambda)|^2}{\bar{S}_{nn}(\Omega; \lambda)}. \quad (110)$$

For a narrowband signal whose response remains fixed, minimizing the noise spectrum in the signal band maximizes the Fisher information. For a broadband waveform, each frequency is weighted by $|s(\Omega; \lambda)|^2 / \bar{S}_{nn}(\Omega; \lambda)$. A setting that gives the smallest spectrum at one frequency can therefore differ from the setting that minimizes the total waveform estimation error.

Equations (102) and (110) assume that the noise covariance is locally independent of θ . A parameter dependent covariance carries additional information and contributes an additional term to the Gaussian Fisher information. Its treatment requires differentiating the complete mean and covariance model.

Replacing the scalar amplitude θ by a time dependent waveform promotes the Fisher information from a number to an integral kernel. The corresponding functional quantum Cramér–Rao bound constrains the covariance of the estimated waveform. Quantum noise cancellation and smoothing can approach that bound in continuous force sensing models [36].

The statistical and continuous SQLs meet at this estimator level. Independent probes contribute Fisher information by summation over resources, whereas a stationary continuous record contributes Fisher information by integration over frequency. Both descriptions ultimately constrain the variance of a calibrated estimator under a stated resource model.

Practical rule of thumb

Use the QFI bound associated with the stated generator and resource count as the separable probe benchmark. Use the matched filter Fisher information for a known waveform in stationary Gaussian noise. Preserve the calibrated response, complete noise covariance, estimator bandwidth, bias assumptions, and resource accounting throughout the comparison.

6 Physical routes below the standard benchmark

Review the meter covariance matrix in Eq. (6), the added noise expression in Eq. (7), the uncorrelated SQL in Eq. (28), and the correlation-aware bounds in Eqs. (50) and (51).

A measurement can move below a standard benchmark by changing the probe covariance, the output quadrature, the measured observable, or the symmetry required of an amplifier. These changes preserve the underlying commutation relations. They alter the assumptions that produced the original benchmark and therefore require a comparison with the limit appropriate to the revised measurement model.

6.1 Squeezed quadratures

For an optical or microwave bosonic mode with annihilation operator $\hat{a}$, define

$$\hat{X}_1 = \frac{\hat{a} + \hat{a}^\dagger}{2}, \quad \hat{X}_2 = \frac{\hat{a} - \hat{a}^\dagger}{2i}. \quad (111)$$

The bosonic relation $[\hat{a}, \hat{a}^\dagger] = 1$ gives

$$[\hat{X}_1, \hat{X}_2] = \frac{i}{2}, \quad \text{Var}(\hat{X}_1) \text{Var}(\hat{X}_2) \geq \frac{1}{16}. \quad (112)$$

Vacuum and coherent states have $\text{Var}(\hat{X}_1) = \text{Var}(\hat{X}_2) = 1/4$.

Introduce the quadrature vector

$$\hat{\mathbf{X}} = \begin{pmatrix} \hat{X}_1 \\ \hat{X}_2 \end{pmatrix}, \quad (113)$$

and define its symmetrized covariance matrix by

$$V_{jk} = \frac{1}{2} \langle \Delta \hat{X}_j \Delta \hat{X}_k + \Delta \hat{X}_k \Delta \hat{X}_j \rangle, \quad \Delta \hat{X}_j = \hat{X}_j - \langle \hat{X}_j \rangle. \quad (114)$$

A pure squeezed Gaussian state with squeeze strength $r_s \geq 0$ and squeeze angle ϕ has covariance

$$\mathbf{V}(r_s, \phi) = \frac{1}{4} \mathbf{R}(\phi) \begin{pmatrix} e^{-2r_s} & 0 \\ 0 & e^{2r_s} \end{pmatrix} \mathbf{R}^\top(\phi), \quad (115)$$

where

$$\mathbf{R}(\phi) = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix}. \quad (116)$$

Its determinant is

$$\det \mathbf{V} = \frac{1}{16}, \quad (117)$$

so the reduction of one quadrature variance is accompanied by the reciprocal increase of the conjugate variance.

A homodyne detector set to angle ζ measures

$$\hat{X}_\zeta = \hat{X}_1 \cos \zeta + \hat{X}_2 \sin \zeta. \quad (118)$$

Using Eq. (115), its variance is

$$\text{Var}(\hat{X}_\zeta) = \frac{1}{4} \left[e^{-2r_s} \cos^2(\zeta - \phi) + e^{2r_s} \sin^2(\zeta - \phi) \right]. \quad (119)$$

The minimum variance $e^{-2r_s}/4$ occurs at $\zeta = \phi$, and the conjugate angle $\zeta = \phi + \pi/2$ has variance $e^{2r_s}/4$. Equation (119) therefore identifies which component of the field is improved by a chosen squeeze angle.

For a conventional interferometric position meter, phase fluctuations usually dominate high frequency readout imprecision, whereas amplitude fluctuations drive radiation pressure backaction at lower frequency. A fixed phase-squeezed input reduces the first contribution and increases the second. A frequency dependent squeeze angle $\phi(\Omega)$ rotates the reduced variance between the quadratures as the relative importance of imprecision and backaction changes [11, 24, 25].

A phase-insensitive loss channel with total efficiency η mixes the state with vacuum,

$$\mathbf{V}_{\text{obs}} = \eta \mathbf{V}(r_s, \phi) + (1 - \eta) \frac{\mathbf{I}}{4}. \quad (120)$$

For detection aligned with the squeezed direction,

$$V_{\text{sq,obs}} = \frac{1}{4} \left[1 - \eta + \eta e^{-2r_s} \right]. \quad (121)$$

Loss therefore imposes a vacuum floor that cannot be removed by increasing r_s .

Small squeeze-angle fluctuations also mix antisqueezing into the measured quadrature. If $\zeta - \phi = \delta\phi$, $\langle \delta\phi \rangle = 0$, and $\sigma_\phi^2 = \langle \delta\phi^2 \rangle \ll 1$, then

$$\text{Var}(\hat{X}_\zeta) \simeq \frac{1}{4} \left[e^{-2r_s} + \left(e^{2r_s} - e^{-2r_s} \right) \sigma_\phi^2 \right]. \quad (122)$$

The factor e^{2r_s} explains why large antisqueezing makes strong squeezing sensitive to small phase errors. Propagation loss, mode mismatch, photodetection efficiency, filter-cavity loss, and angle stability jointly determine the observed reduction.

6.2 Correlation engineering and variational readout

The added displacement noise is

$$\bar{S}_{xx}^{\text{add}} = \bar{S}_{xx}^{\text{imp}} + |\chi_m|^2 \bar{S}_{FF}^{\text{ba}} + 2 \text{Re} \left[\chi_m^* \bar{S}_{xF} \right]. \quad (123)$$

A useful cross spectrum has a phase for which

$$\text{Re} \left[\chi_m^* \bar{S}_{xF} \right] < 0. \quad (124)$$

The negative cross term then removes part of the backaction contribution from the calibrated record.

Assuming $\bar{S}_{FF}^{\text{ba}} > 0$, define the complex correlation ratio

$$c_{xF}(\Omega) = \frac{\bar{S}_{xF}(\Omega)}{\bar{S}_{FF}^{\text{ba}}(\Omega)}. \quad (125)$$

Completing the square gives

$$\begin{aligned} \bar{S}_{xx}^{\text{add}} &= \left(\bar{S}_{xx}^{\text{imp}} - \frac{|\bar{S}_{xF}|^2}{\bar{S}_{FF}^{\text{ba}}} \right) + \bar{S}_{FF}^{\text{ba}} \left| \chi_m + \frac{\bar{S}_{xF}}{\bar{S}_{FF}^{\text{ba}}} \right|^2 \\ &= \left(\bar{S}_{xx}^{\text{imp}} - \frac{|\bar{S}_{xF}|^2}{\bar{S}_{FF}^{\text{ba}}} \right) + \bar{S}_{FF}^{\text{ba}} |\chi_m + c_{xF}|^2. \end{aligned} \quad (126)$$

The first term is the imprecision that remains after its linearly predictable part has been associated with the backaction force. The second term is the residual displacement generated by backaction after correlation-based cancellation.

The determinant inequality in Eq. (18) gives

$$\bar{S}_{xx}^{\text{imp}} - \frac{|\bar{S}_{xF}|^2}{\bar{S}_{FF}^{\text{ba}}} = \frac{\bar{S}_{xx}^{\text{imp}} \bar{S}_{FF}^{\text{ba}} - |\bar{S}_{xF}|^2}{\bar{S}_{FF}^{\text{ba}}} \geq \frac{\hbar^2}{4\bar{S}_{FF}^{\text{ba}}}. \quad (127)$$

Substitution into Eq. (126) yields

$$\bar{S}_{xx}^{\text{add}} \geq \frac{\hbar^2}{4\bar{S}_{FF}^{\text{ba}}} + \bar{S}_{FF}^{\text{ba}} \left| \chi_m + \frac{\bar{S}_{xF}}{\bar{S}_{FF}^{\text{ba}}} \right|^2. \quad (128)$$

The formal choice

$$\frac{\bar{S}_{xF}}{\bar{S}_{FF}^{\text{ba}}} = -\chi_m \quad (129)$$

would cancel the second term. Equation (128) alone does not prove that this cross spectrum can be produced by a causal, stable detector. A physical realization must also satisfy the tighter finite frequency relation in Eq. (17), the associated phase condition, and the correlation-aware bounds in Eqs. (50) and (51) [3, 20].

Variational readout engineers the required correlation by selecting a frequency dependent output quadrature. A minimal lossless input-output model can be written

$$\begin{pmatrix} \hat{X}_1^{\text{out}} \\ \hat{X}_2^{\text{out}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \mathcal{K}(\Omega) & 1 \end{pmatrix} \begin{pmatrix} \hat{X}_1^{\text{in}} \\ \hat{X}_2^{\text{in}} \end{pmatrix} + \begin{pmatrix} 0 \\ \mathcal{G}_x(\Omega)x_{\text{sig}}(\Omega) \end{pmatrix}. \quad (130)$$

Here $\mathcal{K}$ represents the ponderomotive shear that transfers amplitude fluctuations into the phase quadrature, and $\mathcal{G}_x$ is the displacement signal gain. A homodyne detector measures

$$\hat{I}_\zeta = \hat{X}_1^{\text{out}} \cos \zeta + \hat{X}_2^{\text{out}} \sin \zeta. \quad (131)$$

Substitution of Eq. (130) gives

$$\begin{aligned} \hat{I}_\zeta &= \hat{X}_2^{\text{in}} \sin \zeta + [\cos \zeta + \mathcal{K}(\Omega) \sin \zeta] \hat{X}_1^{\text{in}} \\ &\quad + \mathcal{G}_x(\Omega)x_{\text{sig}}(\Omega) \sin \zeta. \end{aligned} \quad (132)$$

In the idealized basis where $\mathcal{K}$ is real, the choice

$$\cot \zeta(\Omega) = -\mathcal{K}(\Omega) \quad (133)$$

removes the amplitude-noise term while retaining a nonzero signal response. A realistic system has optical loss, complex transfer functions, restricted homodyne angles, and dynamical backaction. Its optimum angle follows from the complete calibrated covariance matrix instead of Eq. (133). Broadband experiments have used this output correlation to reduce displacement and force noise below the ordinary uncorrelated SQL [21, 37].

Coherent quantum noise cancellation acts earlier in the measurement chain. Suppose the sensor and an auxiliary system respond to the same backaction force according to

$$\hat{y}_{\text{ba}}(\Omega) = [\chi_m(\Omega) + \mathcal{C}_a(\Omega)\chi_a(\Omega)] \hat{F}_{\text{ba}}(\Omega), \quad (134)$$

where χ_a is the auxiliary susceptibility and $\mathcal{C}_a$ contains its relative coupling and readout gain. Backaction cancels when

$$\mathcal{C}_a(\Omega)\chi_a(\Omega) = -\chi_m(\Omega) \quad (135)$$

over the desired band. A force acting on the sensor alone still produces $\chi_m F_{\text{sig}}$, so the signal remains. The auxiliary system also introduces its own fluctuations, and imperfect matching leaves residual backaction. Negative effective mass responses and quantum-mechanics-free subsystems provide physical routes toward the required opposite susceptibility [38].

6.3 Quantum nondemolition observables and backaction evasion

A quantum nondemolition measurement is designed so that the measured observable remains compatible with its later values. For sampling times t_i and t_j , the ideal condition is

$$[\hat{A}(t_i), \hat{A}(t_j)] = 0 \quad \text{for all relevant } i, j. \quad (136)$$

The Heisenberg equation under $\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}}$ is

$$\frac{d\hat{A}}{dt} = \frac{i}{\hbar}[\hat{H}_0, \hat{A}] + \frac{i}{\hbar}[\hat{H}_{\text{int}}, \hat{A}]. \quad (137)$$

The sufficient interaction condition

$$[\hat{A}, \hat{H}_{\text{int}}] = 0 \quad (138)$$

removes direct meter-induced evolution of $\hat{A}$. Equation (136) additionally requires the free dynamics and sampling protocol to preserve compatibility between successive values. These two conditions distinguish absence of direct measurement disturbance from full repeatability [8, 9, 14].

For a mechanical mode with annihilation operator $\hat{b}$, define rotating quadratures

$$\hat{X}_\varphi = \frac{\hat{b}e^{-i\varphi} + \hat{b}^\dagger e^{i\varphi}}{\sqrt{2}}, \quad \hat{P}_\varphi = \frac{\hat{b}e^{-i\varphi} - \hat{b}^\dagger e^{i\varphi}}{i\sqrt{2}}, \quad (139)$$

which satisfy

$$[\hat{X}_\varphi, \hat{P}_\varphi] = i. \quad (140)$$

Within the rotating wave approximation, a balanced two-tone drive produces an effective interaction of the form

$$\hat{H}_{\text{BAE}} = -\hbar G \hat{X}_c \hat{X}_\varphi, \quad (141)$$

where $\hat{X}_c$ is a cavity-field quadrature. Since

$$[\hat{X}_\varphi, \hat{H}_{\text{BAE}}] = 0, \quad (142)$$

the interaction does not directly disturb the measured mechanical quadrature. Its conjugate evolves according to

$$\left. \frac{d\hat{P}_\varphi}{dt} \right|_{\text{ba}} = G \hat{X}_c, \quad (143)$$

so the measurement backaction is directed into $\hat{P}_\varphi$.

This ideal separation holds in the rotating frame. Counterrotating interactions couple the backaction into $\hat{X}_\varphi$, mechanical damping introduces bath fluctuations, and imbalance between the two pumps opens an additional disturbance channel. Detuning errors rotate the measured quadrature during

acquisition. Strong sideband resolution and balanced, phase-stable drives therefore determine how closely the experiment approaches the ideal interaction [39–41].

The origin of the free-mass position SQL can be seen directly from its Heisenberg motion. For $t_2 > t_1$,

$$\hat{x}(t_2) = \hat{x}(t_1) + \frac{t_2 - t_1}{m} \hat{p}(t_1) + \frac{1}{m} \int_{t_1}^{t_2} (t_2 - t) \hat{F}(t) dt. \quad (144)$$

If a position measurement produces a momentum kick $\Delta \hat{p}_{\text{ba}}$ at t_1 , then

$$\hat{p}(t_1^+) = \hat{p}(t_1^-) + \Delta \hat{p}_{\text{ba}}, \quad (145)$$

and the later position acquires the backaction contribution

$$\Delta \hat{x}_{\text{ba}}(t_2) = \frac{t_2 - t_1}{m} \Delta \hat{p}_{\text{ba}}. \quad (146)$$

The disturbance is therefore carried into every later position estimate.

A speed meter uses repeated interactions to infer a velocity-related observable, which can be represented over an interval τ by

$$\hat{v}_\tau(t) = \frac{\hat{x}(t + \tau) - \hat{x}(t)}{\tau}. \quad (147)$$

Suitable interaction geometry makes the leading low-frequency backaction contributions enter the output with opposite signs. Contractive-state protocols and realizable generalized measurements likewise show that a free-mass SQL depends on the prepared state, measured observable, and temporal measurement protocol [42, 43]. Equation (34) should therefore be identified specifically as the SQL of the stated uncorrelated continuous position meter.

6.4 Phase-preserving amplification

A phase-preserving amplifier amplifies signals of arbitrary phase by applying the same amplitude gain to both field quadratures. Let $G \geq 1$ denote its power gain, so that $\sqrt{G}$ is the corresponding amplitude gain. A general linear input-output relation can be written

$$\hat{a}_{\text{out}} = \sqrt{G} \hat{a}_{\text{in}} + \hat{L}^\dagger, \quad G \geq 1, \quad (148)$$

where $\hat{L}$ acts on internal amplifier degrees of freedom and commutes with the input mode,

$$[\hat{L}, \hat{a}_{\text{in}}] = [\hat{L}, \hat{a}_{\text{in}}^\dagger] = 0. \quad (149)$$

The output must remain a valid bosonic mode, so its commutator must satisfy

$$\begin{aligned} [\hat{a}_{\text{out}}, \hat{a}_{\text{out}}^\dagger] &= G[\hat{a}_{\text{in}}, \hat{a}_{\text{in}}^\dagger] + [\hat{L}^\dagger, \hat{L}] \\ &= G + [\hat{L}^\dagger, \hat{L}] = 1. \end{aligned} \quad (150)$$

Consequently,

$$[\hat{L}, \hat{L}^\dagger] = G - 1. \quad (151)$$

Any amplifier with $G > 1$ therefore requires an additional quantum degree of freedom whose fluctuations enter the output together with the amplified signal.

The minimum-noise single-mode realization represents this degree of freedom by an ancillary bosonic mode $\hat{b}_{\text{in}}$ satisfying

$$[\hat{b}_{\text{in}}, \hat{b}_{\text{in}}^\dagger] = 1. \quad (152)$$

Setting

$$\hat{L} = \sqrt{G-1} \hat{b}_{\text{in}} \quad (153)$$

gives the canonical phase-preserving amplification map

$$\boxed{\hat{a}_{\text{out}} = \sqrt{G} \hat{a}_{\text{in}} + \sqrt{G-1} \hat{b}_{\text{in}}^\dagger}. \quad (154)$$

Equation (154) defines the amplification transformation itself. For a zero-mean ancillary mode, $\langle \hat{b}_{\text{in}} \rangle = 0$, it gives

$$\langle \hat{a}_{\text{out}} \rangle = \sqrt{G} \langle \hat{a}_{\text{in}} \rangle. \quad (155)$$

The complex field amplitude is therefore multiplied by the real factor $\sqrt{G}$ without changing its phase.

The creation operator in Eq. (154) contributes with the opposite commutator sign from the signal annihilation operator,

$$\begin{aligned} [\hat{a}_{\text{out}}, \hat{a}_{\text{out}}^\dagger] &= G[\hat{a}_{\text{in}}, \hat{a}_{\text{in}}^\dagger] + (G-1)[\hat{b}_{\text{in}}^\dagger, \hat{b}_{\text{in}}] \\ &= G - (G-1) = 1. \end{aligned} \quad (156)$$

The ancillary contribution is therefore required for commutator preservation and cannot be removed from a deterministic phase-preserving amplifier.

Using the quadrature normalization introduced in Eq. (111), define the ancillary quadratures by

$$\hat{X}_{1,b} = \frac{\hat{b}_{\text{in}} + \hat{b}_{\text{in}}^\dagger}{2}, \quad \hat{X}_{2,b} = \frac{\hat{b}_{\text{in}} - \hat{b}_{\text{in}}^\dagger}{2i}. \quad (157)$$

Equation (154) then becomes

$$\hat{X}_{1,\text{out}} = \sqrt{G} \hat{X}_{1,\text{in}} + \sqrt{G-1} \hat{X}_{1,b}, \quad (158)$$

and

$$\hat{X}_{2,\text{out}} = \sqrt{G} \hat{X}_{2,\text{in}} - \sqrt{G-1} \hat{X}_{2,b}. \quad (159)$$

Both input quadratures receive the same amplitude gain $\sqrt{G}$. The sign difference in the ancillary terms preserves the output quadrature commutator,

$$[\hat{X}_{1,\text{out}}, \hat{X}_{2,\text{out}}] = \frac{i}{2}. \quad (160)$$

Equal gain for both signal quadratures is the defining feature of phase-preserving amplification.

Assume that the input and ancillary modes are uncorrelated. Equations (158) and (159) then give

$$\text{Var}(\hat{X}_{j,\text{out}}) = G \text{Var}(\hat{X}_{j,\text{in}}) + (G-1) \text{Var}(\hat{X}_{j,b}), \quad j \in \{1, 2\}. \quad (161)$$

Dividing by the power gain refers the output variance back to the input,

$$\frac{\text{Var}(\hat{X}_{j,\text{out}})}{G} = \text{Var}(\hat{X}_{j,\text{in}}) + \left(1 - \frac{1}{G}\right) \text{Var}(\hat{X}_{j,b}). \quad (162)$$

The first term is the original input fluctuation. The second is the input-referred noise contributed by the ancillary mode.

For an ancillary thermal state with mean occupation $\bar{n}_b$,

$$\text{Var}(\hat{X}_{j,b}) = \frac{2\bar{n}_b + 1}{4}, \quad j \in \{1, 2\}. \quad (163)$$

The variance added to either input-referred quadrature is therefore

$$V_{j,\text{add}} = \left(1 - \frac{1}{G}\right) \frac{2\bar{n}_b + 1}{4}. \quad (164)$$

With the quadrature normalization used here, define the symmetrized input-referred added-noise number by

$$n_{\text{add}} = 2V_{j,\text{add}}. \quad (165)$$

This convention expresses the added per-quadrature variance as an equivalent bosonic noise occupation. Equations (164) and (165) give

$$n_{\text{add}} = \left(1 - \frac{1}{G}\right) \left(\bar{n}_b + \frac{1}{2}\right). \quad (166)$$

Since every physical ancillary state satisfies $\bar{n}_b \geq 0$, every deterministic phase-preserving amplifier described by Eq. (154) obeys the finite-gain input-referred added-noise bound

$$\boxed{n_{\text{add}} \geq \frac{1}{2} \left(1 - \frac{1}{G}\right)}. \quad (167)$$

Equation (167) quantifies the unavoidable noise cost of phase-preserving gain. It is a lower bound on the noise added by the transformation in Eq. (154), rather than the transformation itself.

Vacuum in the ancillary mode has $\bar{n}_b = 0$ and reaches equality in Eq. (167). The high-gain limit is

$$\lim_{G \rightarrow \infty} n_{\text{add}} = \frac{1}{2}. \quad (168)$$

The familiar half quantum therefore follows from three requirements: deterministic gain, equal amplification of both quadratures, and preservation of the bosonic commutator. Any thermal occupation of the ancillary mode raises the added noise above this minimum [12, 44].

An ideal phase-sensitive amplifier changes the measurement symmetry by applying reciprocal gains to two selected quadratures,

$$\hat{X}_{1,\text{out}} = \sqrt{G} \hat{X}_{1,\text{in}}, \quad \hat{X}_{2,\text{out}} = \frac{1}{\sqrt{G}} \hat{X}_{2,\text{in}}. \quad (169)$$

The quadrature commutator remains unchanged,

$$[\hat{X}_{1,\text{out}}, \hat{X}_{2,\text{out}}] = [\hat{X}_{1,\text{in}}, \hat{X}_{2,\text{in}}] = \frac{i}{2}, \quad (170)$$

without introducing an auxiliary noise mode. The selected quadrature can therefore be amplified without added noise in the ideal unitary model, as the conjugate quadrature is deamplified. This transformation does not give equal gain to signals of arbitrary phase, so it lies outside the assumptions that produce the phase-preserving half-quantum limit.

Table 3: Representative strategies for surpassing a standard benchmark.

Method	Change to the measurement model	Representative platform
Phase squeezing	Reduces detected phase quadrature variance	Optical phase sensors and interferometers
Frequency dependent squeezing	Rotates the noise ellipse across Fourier frequency	Gravitational wave detectors

Method	Change to the measurement model	Representative platform
Variational readout	Selects a correlated output quadrature that cancels backaction in the estimator	Interferometers and optomechanics
Backaction evasion	Couples the meter to one rotating mechanical quadrature	Microwave and optical optomechanics
Speed meter	Measures a velocity like observable with repeated interaction geometry	Gravitational wave detector concepts
Coherent noise cancellation	Matches an auxiliary response of opposite sign to the sensor response	Optomechanical force sensors
Spin squeezing	Reduces collective spin variance at fixed mean spin signal	Atomic clocks and magnetometers
Entangled probes	Increases total generator variance above separable scaling	Optical and atomic phase estimation
QND measurement	Selects an observable compatible with its future values	Oscillators, photons, spins, and qubits
Phase sensitive amplification	Amplifies one chosen field quadrature	Microwave qubit readout and radio frequency sensing

Practical rule of thumb

Identify the assumption changed by each sub-SQL technique. Squeezing changes the probe covariance, variational readout changes the measured quadrature, QND changes the observable, and entanglement changes generator variance.

7 Applications and evidence standards

Review the difference among imprecision, added noise, and total measured noise. A claim about one level gives limited information about the others.

7.1 Gravitational wave interferometry

Laser interferometers infer strain from differential arm motion over kilometer baselines. Optical phase noise dominates much of the high frequency quantum noise, and radiation pressure motion gains importance at lower frequencies. Squeezed vacuum injection first improved detector sensitivity beyond the coherent shot noise level, Advanced LIGO later operated with squeezed vacuum, and frequency dependent squeezing then provided broadband quantum noise reduction [26, 45–47].

The LIGO Livingston analysis reported the quantum noise contribution below the free mass SQL from 35 to 75 Hz with a maximum improvement of 3 dB near 50 Hz after installation of the A+ frequency dependent squeezing system [48, 49]. Classical noise exceeded quantum noise in that low frequency band, so the reported quantum spectrum was inferred by calibrated model subtraction and dedicated squeeze angle measurements. The distinction preserves the importance of the result and states its experimental scope accurately.

7.2 Optomechanical and atomic force sensing

These experiments mark three distinct stages in the development of quantum limited force sensing. Murch and collaborators directly observed radiation pressure backaction, establishing the meter force as a measurable physical contribution [28]. Schreppler and collaborators then resolved the tradeoff between imprecision and backaction in a calibrated force measurement [29]. Mason and collaborators used the resulting correlation constructively, reducing the total off resonance force and displacement noise below the uncorrelated SQL after thermal motion and detector noise had been included [21].

In atomic force microscopy, the force estimate is inferred from cantilever motion, so cantilever thermal fluctuations and optical readout noise enter the same calibrated record. Increasing optical power can reduce imprecision, yet it can also increase radiation pressure and local heating. The tip sample interaction adds another force channel that is absent from a free cantilever calibration [50, 51]. Pooser and collaborators used truncated nonlinear interferometry to reduce the optical readout noise below the coherent reference and reached $1.7 \text{ fm}/\sqrt{\text{Hz}}$ [52]. That result establishes quantum enhanced optical readout. A total force SQL test also requires calibrated thermal motion, backaction, and tip sample force noise over the same measurement band.

Magnetic resonance force microscopy reached single electron spin detection by coupling spin dynamics to a sensitive cantilever [53]. That milestone concerns signal detection and force sensitivity. Its relation to an SQL requires the complete quantum and thermal noise budget for the chosen force estimator.

7.3 Spin ensembles and clocks

For N uncorrelated two level atoms polarized along one collective spin direction, transverse projection noise scales as $\sqrt{N}$ and phase uncertainty scales as $N^{-1/2}$. The Wineland metrological squeezing parameter is

$$\xi_R^2 = \frac{N(\Delta J_\perp)^2}{|\langle \mathbf{J} \rangle|^2}. \quad (171)$$

Values $\xi_R^2 < 1$ indicate phase sensitivity below the coherent spin state SQL after contrast loss is included. The parameter connects reduced collective variance with a usable signal slope and remains central to atomic clocks, magnetometers, and atom interferometers [5, 54, 55].

7.4 Spatial parameter estimation and quantum imaging

In quantum imaging, the unknown quantity may be a source position, source separation, object displacement, spatial phase, or another parameter encoded in the optical field. Diffraction determines which spatial modes reach the detector, and estimation theory determines how accurately the parameter can be inferred from those modes. For a single spatial coordinate r , the quantum Cramér–Rao bound gives

$$\text{Var}(\hat{r}) \geq \frac{1}{\nu F_Q(r)}. \quad (172)$$

For several spatial parameters, the corresponding quantum Fisher information matrix determines the covariance bounds and reveals any incompatibility among their optimal measurements.

Spatial mode measurements retain sensitivity to source separation because the parameter changes the modal composition of the field even after the intensity profile varies only weakly. A squeezed or number correlated probe can add a second advantage by reducing fluctuations in the modes that carry the parameter derivative. Loss and detector inefficiency mix those informative modes with vacuum and

reduce the gain, so the benchmark must use the same photon budget and receiver measurement as the experiment [56].

In active imaging, nonclassical spatial modes can reduce uncertainty in beam displacement and optical phase estimates by concentrating correlations in the modes that carry the parameter response. The attainable gain depends on how well propagation and detection preserve those correlations, so comparisons should hold photon number, state-preparation cost, and receiver measurement fixed. The resulting advantage may originate from the probe state, the modal measurement, or both, and should be evaluated against matched classical and quantum estimation bounds.

Practical rule of thumb

When evaluating a published, proposed, or experimentally reported superresolution result, the analysis should state the estimated parameter, optical state, measurement strategy, photon budget, and reference bound. A comparison with the Rayleigh criterion evaluates resolution under a specified imaging procedure, whereas comparison with the quantum Cramér–Rao bound evaluates the estimator precision permitted by the probe state, propagation channel, and available measurement.

7.5 A hierarchy of experimental claims

Table 4: Evidence required for increasingly strong SQL statements.

Claim	Required comparison	Common missing element
Sub shot noise readout	Detected readout variance against a coherent or vacuum reference at equal signal resource	Loss corrected resource accounting
Imprecision below SQL scale	Calibrated input referred imprecision against a stated SQL displacement scale	Backaction and thermal noise
Quantum noise below SQL	Calibrated sum of quantum imprecision, backaction, and correlations against the same SQL convention	Classical noise and model uncertainty
Total sensitivity below SQL	Measured total estimator noise, including thermal and technical contributions, against a calibrated SQL curve	Independent calibration and uncertainty propagation
Metrological gain below separable bound	Estimator variance or Fisher information against equal resource separable probes	State preparation cost, contrast, and bias

Table 5: How the SQL enters representative platforms.

Platform	Measured parameter	Standard balance or resource restriction	References
Laser interferometer	Differential displacement or strain	Phase imprecision against radiation pressure backaction	[11, 26, 48]
Cavity optomechanics	Displacement or force	Optical imprecision against photon number backaction	[16, 21]
Atomic force microscope	Cantilever motion and force	Optical readout, cantilever disturbance, and thermal force	[50, 52]
Spin ensemble	Phase, frequency, or field	Independent projection noise against collective correlations	[54, 55]
Microwave amplifier	Field quadratures	Phase-preserving gain against commutator-preserving added noise	[12, 44]
Quantum phase estimation	Optical or atomic phase	Separable generator variance against correlated generator variance	[30, 33]
Quantum imaging	Source position, separation, or another spatial parameter	Direct imaging compared with quantum-optimal spatial-mode measurements; nonclassical-probe bounds require a separately specified active-imaging model	[18, 56]

Practical rule of thumb

Reserve the phrase total sensitivity below the SQL for a comparison that includes intrinsic mechanical motion, technical noise, measurement imprecision, backaction, correlations, and calibration uncertainties in the same units and bandwidth.

8 Experimental design and numerical verification

Before beginning the numerical analysis, specify the parameter to be estimated, the estimator bandwidth, and the spectral convention. These choices define the physical problem and provide the basis for a meaningful optimization.

8.1 Noise budget and calibration sequence

A defensible SQL comparison can be built with the following sequence.

1. Define the input variable and estimator. State whether the result refers to displacement, force, strain, phase, frequency, or field.
2. Measure or model the complex transfer function from the input variable to the recorded channel. Preserve its phase.
3. Fix the Fourier normalization, spectral sidedness, symmetrization, resolution bandwidth, windowing, and averaging method.
4. Calibrate imprecision with the physical coupling suppressed or independently varied.
5. Calibrate backaction by varying probe strength and measuring the driven response or induced heating.
6. Estimate the cross spectrum with phase information. A magnitude squared coherence by itself discards the sign needed for cancellation.
7. Measure thermal and technical backgrounds and propagate their calibration uncertainty.
8. Construct the SQL from the measured effective susceptibility and the same convention used for data.
9. Repeat the comparison across probe strengths and readout quadratures to show the expected covariance geometry.

For a total measured displacement spectrum,

$$\bar{S}_{xx}^{\text{meas}} = \bar{S}_{xx}^{\text{intrinsic}} + \bar{S}_{xx}^{\text{tech}} + \bar{S}_{xx}^{\text{add}}. \quad (173)$$

The intrinsic spectrum can be decomposed into thermal and zero-point contributions when the adopted noise model treats them as separate terms. In some analyses, modeled classical backgrounds are subtracted from the measured spectrum to infer the remaining quantum-noise contribution, as occurs in low-frequency interferometer studies. Any conclusion that this inferred quantum component reaches or falls below a stated SQL should specify the background-subtraction procedure, the assumptions entering the noise model, and the uncertainty propagated into the inferred spectrum.

8.2 Control parameters for an SQL heat map

The analytic heat map in Fig. 4 has clear controls. The damping ratio γ sets resonance width, the plotted frequency interval sets the mechanical response range, the range of $\log_{10} \lambda$ sets the available measurement strength, and the sample count sets rendering density. These values appear as control knobs in the preamble. A laboratory version can replace Eq. (42) with measured complex susceptibility and replace Eq. (44) with the measured covariance matrix.

Define a dimensionless excess added noise metric

$$H(\Omega, \lambda) = \log_{10} \left[\frac{\bar{S}_{xx}^{\text{add}}(\Omega, \lambda)}{\bar{S}_{xx}^{\text{SQL}}(\Omega)} \right]. \quad (174)$$

$H = 0$ lies on the selected SQL, $H > 0$ lies above it, and $H < 0$ lies below it. A plot with correlations must evaluate the complex cross term in Eq. (7). Using $|\bar{S}_{xF}|$ without its phase can produce a physically incorrect valley.

8.3 Common diagnostic failures

Below are several checks to catch common errors before a sub-SQL conclusion is drawn.

- Dimensional consistency must hold after input referral. A force spectrum multiplied by $|\chi_m|^2$ has displacement spectrum units.
- The optimum of Eq. (22) must occur where its two positive terms are equal.
- Conversion from a double-sided to a one sided power spectrum multiplies by two, and the corresponding amplitude spectrum multiplies by $\sqrt{2}$.
- A squeezed variance quoted in decibels uses $10 \log_{10}(V/V_0)$. An amplitude ratio uses $20 \log_{10}(A/A_0)$.
- A sub-SQL quantum component can coexist with total detector noise above the SQL.
- A narrowband noise minimum can differ from the setting that maximizes broadband waveform Fisher information.

Practical rule of thumb

Treat susceptibility phase, cross spectral phase, sidedness, and calibration covariance as primary data. These quantities decide whether a plotted sub-SQL valley represents physical sensitivity or a convention mismatch.

9 Synthesis

SQL takes on a definite physical meaning once the measured quantity, available resources, probe state, estimator, bandwidth, and noise convention are specified. For an uncorrelated continuous position measurement, the symmetrized double-sided displacement spectrum reaches $\hbar|\chi_m|$. An ensemble of independent probes follows the familiar $N^{-1/2}$ scaling of statistical uncertainty, whereas a phase-preserving amplifier approaches one-half quantum of input-referred added noise at large gain. Force sensing that exploits imprecision–backaction correlations is subject to a lower dissipative bound set by the imaginary part of the inverse mechanical susceptibility. Although these results share a common foundation in quantum mechanics, each applies to a different measurement task, so a direct comparison requires consistent experimental conditions, definitions, and conventions.

Quantum-enhanced sensing improves precision by changing how quantum uncertainty enters the measurement. Squeezing concentrates fluctuations in the variable that matters most for the estimate, and variational readout uses correlations between imprecision and backaction so that part of the disturbance can be removed from the inferred signal. Backaction-evading schemes and speed meters go further by changing which mechanical quantity is monitored, thereby preventing the measurement disturbance from returning in the same form. Entanglement enlarges the response of the collective probe to the parameter of interest, and QND interactions preserve the measured observable so that it can be interrogated repeatedly. These methods all respect the quantum uncertainty relations, yet they exceed reference limits that assume uncorrelated probes, fixed observables, or simpler measurement couplings.

A convincing demonstration consists of defining the reference curve and the estimator compared with it. The resource count and spectral convention set the numerical scale of the benchmark, and the calibrated noise model determines which fluctuations enter the reported sensitivity. Propagating calibration uncertainty through the same estimator then permits comparisons across otherwise different sensing platforms without conflating readout imprecision with total sensitivity.

Practical rule of thumb

SQL is a calibrated design benchmark. A complete sub-SQL result identifies the assumption that has changed and demonstrates the improvement using the full estimator noise budget.

A Fourier and spectral conversion reference

With

$$A(\Omega) = \int_{-\infty}^{\infty} A(t) e^{i\Omega t} dt, \quad A(t) = \int_{-\infty}^{\infty} \frac{d\Omega}{2\pi} A(\Omega) e^{-i\Omega t}, \quad (175)$$

Eq. (4) gives the double-sided symmetrized spectrum. For a real stationary process,

$$\langle A^2(t) \rangle = \int_{-\infty}^{\infty} \frac{d\Omega}{2\pi} \bar{S}_{AA}(\Omega) = \int_0^{\infty} \frac{d\Omega}{2\pi} S_{AA}^{(1)}(\Omega). \quad (176)$$

Spectra per hertz use $f = \Omega/(2\pi)$ and satisfy $S^{[f]}(f) df = S^{[\Omega]}(\Omega) d\Omega$. Stating the integration measure prevents a hidden factor of 2π .

B Analytic heat map definition

The heat map uses

$$r = \frac{\Omega}{\Omega_m}, \quad \ell = \log_{10} p, \quad \gamma = \frac{\Gamma_m}{\Omega_m}. \quad (177)$$

Its plotted function is

$$H(r, \ell) = \log_{10} \left[\frac{10^{-\ell} + 10^{\ell} |\tilde{\chi}_m(r)|^2}{2 |\tilde{\chi}_m(r)|} \right]. \quad (178)$$

The inequality

$$\frac{1}{2} (u + u^{-1}) \geq 1, \quad u = 10^{\ell} |\tilde{\chi}_m(r)|, \quad (179)$$

gives $H(r, \ell) \geq 0$. Equality occurs along

$$\ell_{\text{opt}}(r) = -\log_{10} |\tilde{\chi}_m(r)|. \quad (180)$$

This analytic construction replaces the external `sql_heatmap.dat` dependency and compiles from the supplied TeX source and bibliography database.

C Compact formula sheet

Table 6: Frequently used formulas and their scope.

Formula	Scope
$\bar{S}_{xx}^{\text{add}} = \bar{S}_{xx}^{\text{imp}} + \chi_m ^2 \bar{S}_{FF}^{\text{ba}} + 2 \text{Re}(\chi_m^* \bar{S}_{xF})$	Linear input referred displacement measurement
$\bar{S}_{xx}^{\text{SQL}} = \hbar \chi_m $	Ideal uncorrelated meter, symmetrized double-sided spectrum
$\bar{S}_{FF}^{\text{SQL}} = \hbar / \chi_m $	Force referred form of the same SQL
$\bar{S}_{FF}^{\text{add}} \geq \hbar \text{Im } \chi_m^{-1} $	Correlation aware stationary dissipative force bound
$F_Q = 4(\Delta G)^2$	Pure state under unitary parameter encoding
$F_Q \leq \sum_j (\Delta g_j)^2$	Separable probes with bounded local generator ranges
$F_Q = N^2$	Ideal N qubit GHZ probe under the stated local generator
$n_{\text{add}} \geq (1 - G^{-1})/2$	Deterministic phase preserving bosonic amplifier
$\xi_R^2 = N(\Delta J_{\perp})^2 / \langle \mathbf{J} \rangle ^2$	Metrological spin squeezing parameter

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